There are three levels of data abstraction:

1) Physical level: Describes how data is stored in database. Users do not interact with this level.

2) Logical level: Defines what data is stored such as patient details, doctor information and appointment & the relationship b/w them.

3) View Level: Show only required information to different users. For example, doctors can see medical history, nurses can see patient details

Ques3. Explore the potential benefits and challenges associated with implementing data abstraction in this context.

Ans Benefits:

1) Improves data security & privacy
2) Makes the system easy to use for healthc staff.
3) Provides data Independence, so changes in database structure do not affect users.
4) Helps in better data management.

Challenges

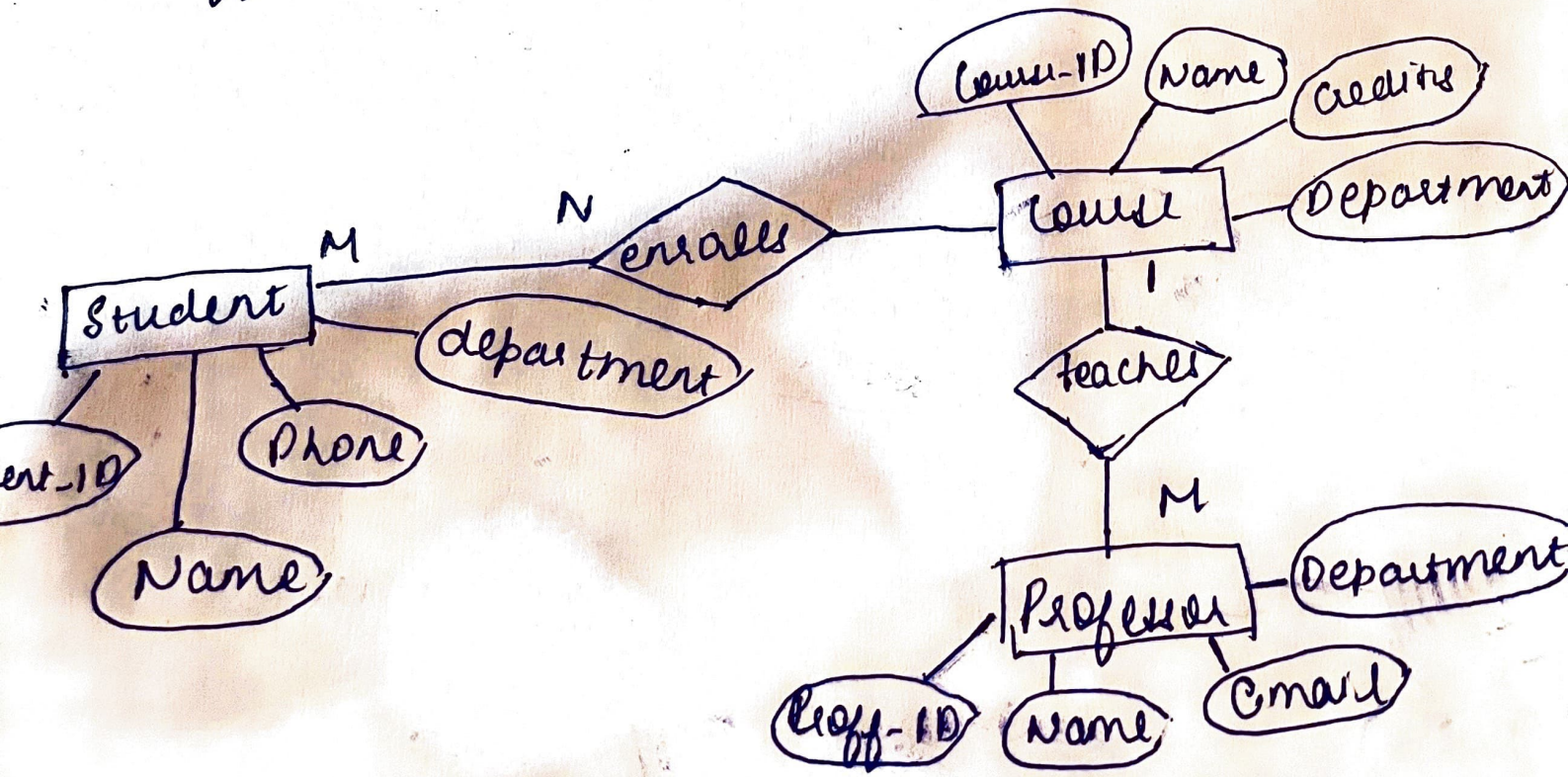
1) Implementation complexity
2) Performance overhead
3) High development cost

curity risks if poorly implemented.
Training requirements

Section-C

Ques 9. Design a database for a university, incorporating entities like students, courses, and professors using the entity-relationship model, illustrate and explain the relationship b/w these entities. Include attributes for each entity to capture relevant information such as student names, course, details and professor information

Ans Student (Student_ID(PK), Name, Email, Phone, Department)
Course (Course_ID(PK), Name, Credits, Department)
Attributes (Staff_ID(PK), Name, Department)
Relationship.
1) Student Enrolls Course m:m
2) Prof - Teaches - Course 1:m



Sec - A

Ques Fill in the blanks

The Entity Relationship Diagram ~~was~~ visually represents how entities relate in a database, offering a clear blueprint of the data's organizational structure.

In the Relational Model data is organized into tables, with relationships b/w entities defined through foreign key constraints

The object oriented model treat data as a collection of objects, each with unique attributes and methods, mirroring real world entities and interactions.

Integrity constraints ensure the reliability of data by enforcing rules that must be adhered to maintain the accuracy of information

Section - B

In a health care system, how data abstract simplify the interaction b/w health care professionals and patient records.

Data abstraction means hiding complex details and showing only important information to users. In a health care system, it helps doctors, nurses and staff access patient records easily